

CS 461 Homework 1

Due: Sept. 30 11:59 pm

1. [ML Principles] Imagine you want to teach a robot the concept of a cup. You show it various objects like cups, cars, lamps, and so on. During the training, the robot uses the condition below to decide if the shown object is a cup or not, and update its black box system accordingly. This condition function works like the loss in machine learning, guiding the robot on how well it is performing.

training condition-1: Cup is an object that can contain liquid

1.1 Do you think the training condition is well-defined for the target task: recognizing a cup? Please explain your reasoning.

1.2 What do you think the robot's performance would be like if it is evaluated using recall or precision on the training samples? Please compare the expected performance of the two metrics.

1.3 Are the precision and recall good metrics for measuring the robot's training performance? If not, how would you design an alternative one?

1.4 If you can change the training condition-1, how would you change it? and how would this change affect the robot's training performance?

1.5 Suppose you train the robot using your new condition. Do you think the robot would recognize the artwork 'Breakfast in Fur' by M  ret Oppenheim as a cup? To answer this question, what factors do you need to consider? For example, what if the robot has not been exposed to a wide range of cup examples or what if the robot's recognition process is based on a mathematical function rather than a human-like perception system like a brain? Please explain your reasoning.

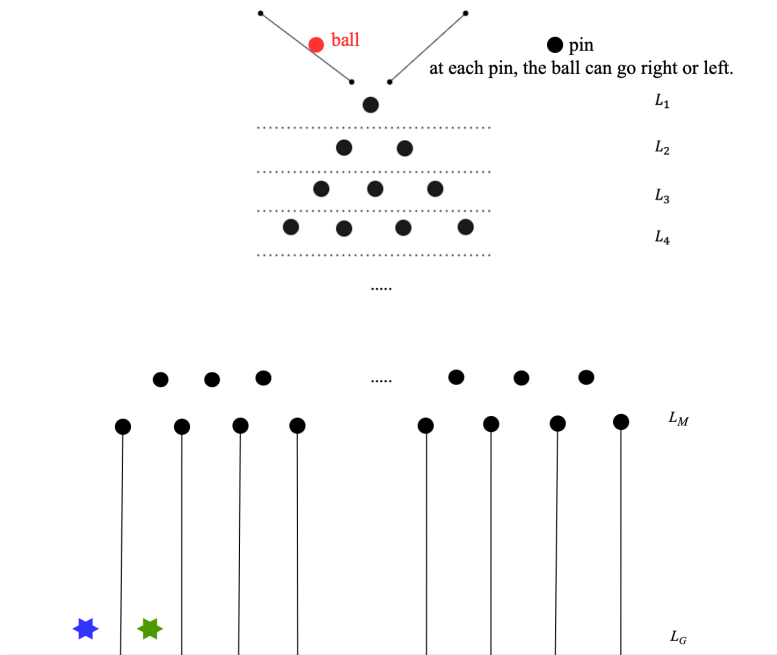
[Breakfast in Fur' by M  ret Oppenheim]



1.6 Based on the answers in the previous problems of 1.1-1.4, how would you set your rules to train the robot? Please describes desirable rules for the following items.

- training condition (loss):
- performance metric:
- number training samples and the range of variation:
- robot recognition process: a simple function, complex one, or something else like an imaginary system simulating a human brain?:

2. [Galton Board] Suppose we have a Galton board as below and run an experiment to drop a ball into it. At each pin, the ball takes a path left or right with equal probability.



2.1 Define the sample spaces $\Omega_1, \Omega_2, \dots, \Omega_M$, where Ω_i is the set of all possible sides the ball can take at the level L_i .

2.2 Define the sample space $\Omega = \Omega_1 \times \Omega_2 \times \dots \times \Omega_M$. Each element of the set Ω encodes a path the ball could take until it arrives at the ground-level L_G .

2.3 What is the meaning of the location at L_G where the ball finally arrives? (hint: think about the possible ball path resulting in the different locations like the leftmost (blue star) or the second left location (green star)).

2.4 How would you represent the location numerically? Please define a random variable that maps Ω to the numerical values.

2.5 Simulate 1,000 ball drops and estimate the Probability Mass Function (PMF) of your random variable for the depth $M = 5$, $M = 10$, $M = 100$. Use `numpy.random.uniform` for your simulation. Plot them and check how your PMFs change as M goes to large. Please explain the phenomenon in relation to Central Limit Theorem.

2.6 In the class we learned a regression model $y = f(x) + \epsilon$ where ϵ follows a Gaussian. Why Gaussian is a reasonable to model the error in the regression modeling?

3. [Bayes Rule] A computer manufacturer uses chips from three sources uniformly. Chips from sources A, B, and C are defective with probabilities 0.001, 0.005, and 0.01, respectively.

3.1 What is the probability of a chip is defective.

3.2 If a randomly selected chip is found to be defective, find the probability that the manufacturer was A.

3.3 If a randomly selected chip is found to be non-defective, find the probability that the manufacturer was C.

3.4 Do you think the defectiveness and manufacturer sources are statistically independent or dependent? Please explain your reasoning.

4. [Naïve Bayes] You will build a Naive Bayes Classifier to recognize diabetes based on the two factors of glucose and blood pressure. Let the random variable D be the indicator of diabetes, and the two random variables G and B represent glucose and blood pressure each. (note: you can use numpy or math but please do not use other libraries like SciPY).

4.1 Naïve Bayes is a probabilistic model based on Bayes Theorem. Rewrite the formula below as G and B are conditionally independent given $D+$. Also, write about $P[D = -|G = g, B = b]$. How would you use the two formulas to determine diabetes when you have a glucose and blood pressure record (g, b) ?

$$P[D = +|G = g, B = b] = \frac{P[D = +, G = g, B = b]}{P[G = g, B = b]} = \frac{P[G = g, B = b|D = +] \cdot P[D = +]}{P[G = g, B = b]}$$

4.2 The data file “train.csv” contains patient records: glucose level, blood pressure, and diagnosis outcomes. Based on the formula you wrote in 4.1, estimate the necessary densities based on “train.csv”. Glucose and blood pressure conditioned on diabetes follow Gaussian densities and use ML (Maximum Likelihood) estimators presented in the textbook Bishop section 1.2.4. Your file outcome will be “yourname-nb-train.py”.

- for example, $f_G(g|D+) = N(\mu_{g-pos}, \sigma_{g-pos}^2)$
- $f_B(b|D-) = N(\mu_{b-neg}, \sigma_{b-neg}^2)$, and so on.

4.3 Write a Naïve Bayes classifier code “yourname-nb-cls.py” based on your estimated densities.

4.4 Evaluate your classifier using “test.csv” and report your test accuracy.

4.5 Do you think the data reflects reality well? Which part of the previous steps would you like to change if we cannot collect the data again? How would you change?

5. [Data Whitening] Whitening data is a key preprocessing in ML. By using “x.npz”, estimate the first and second order statistics and whiten the data. For your computation, you can use `numpy.matmul`, `numpy.sum`, and `numpy.linalg.svd`.

5.1 Estimate $E[X] = \vec{\mu}$ and $COV[X, X] = \Sigma$.

5.2 Let W be a random vector defined by $\vec{W} = A\vec{X} + \vec{b}$. Express $E[W]$ and $COV[W, W]$ in terms of $E[X] = \vec{\mu}$ and $COV[X, X] = \Sigma$.

5.3 Compute A and b to whiten X . i.e. $E[W] = 0$ and $COV[W, W] = I$

6. [Extra 20 Points, ML and MAP Estimation] Suppose we want to measure a length, for example, the water depth μ at $79.137^\circ(\text{N})$ and $2.817^\circ(\text{E})$. The depth was measured repeatedly and recorded as $x_1, x_2, \dots, x_n$. For device imperfection, the samples were varied by ε where $\varepsilon \sim N(0, \sigma^2)$. i.e $x = \mu + \varepsilon$. Bayes rule allows us to evaluate the uncertainty in μ after observing $\vec{x}$ in the posterior probability $p(\mu|\vec{x})$ as below.

$$\mu^* = \arg \max_{\mu} p(\mu|\vec{x}) \propto p(\vec{x}|\mu) \cdot p(\mu)$$

6.1 Given the observations $x_1, x_2, \dots, x_n$, derive a formula to estimate μ_{ML}^* when μ is a fixed value. We assume the observations are i.i.d (independent and identically distributed) and follow multivariate Gaussian.

$$f(\vec{x}) = \frac{1}{(2\pi\sigma^2)^{N/2}} \exp \left\{ \frac{-1}{2\sigma^2} \sum_{n=1}^N (x_n - \mu)^2 \right\}$$

6.2 Given the observations $x_1, x_2, \dots, x_n$, derive a formula to estimate μ_{MAP}^* when μ is known to follow Gaussian $p(\mu) \sim N(\mu_0, \sigma_0^2)$. (hint: the posterior $p(\mu|\vec{X}) \propto p(\vec{X}|\mu) \cdot p(\mu)$ is also Gaussian like $\sim N(\mu_N, \sigma_N^2)$. Then, it will have a mode at μ_N .)

6.3 The μ_{MAP}^* estimated in 6.2 can be expressed as follows. Examine how the behavior of μ_{MAP}^* changes for the two cases $N \rightarrow \infty$ and $N \rightarrow 0$ in relation to μ_0 and μ_{ML}^* . If you have enough data points, what method would you choose between ML and MAP?

$$\mu_{MAP}^* = \frac{\sigma^2}{N\sigma_0^2 + \sigma^2} \cdot \mu_0 + \frac{N\sigma_0^2}{N\sigma_0^2 + \sigma^2} \cdot \mu_{ML}^*$$